

mdstats periodic lattice-gauge specification

Stage C0A1: cell identity, basis continuity, and explicit unimodular reconciliation

mdstats

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Scope

`mdstats.coordinates.periodic_gauge` validates the identity of the reported periodic lattice basis before Stage C0A2 interprets frame-to-frame cell changes as physical affine deformation. It does not construct the affine registration map.

The controlling failure rule is:

a basis relabeling is never silently interpreted as strain

Row-vector lattice convention

Cells follow the package convention

$$\mathbf{x} = \mathbf{f}H.$$

Two reported cells can describe the same lattice under an integer basis change

$$H' = UH, \quad U \in GL(3, \mathbb{Z}), \quad |\det U| = 1.$$

For each frame, the gauge stores an integer matrix Q_t such that

$$\tilde{H}_t = Q_t H_t$$

is the gauged cell used by downstream registration. Without reconciliation, $Q_t = I$ for every accepted frame.

Validation order

For every source cell, the implementation checks:

1. finite 3×3 shape;
2. determinant magnitude above the configured singularity tolerance;
3. condition number below the configured limit;
4. handedness and periodic-axis compatibility; and
5. basis/deformation continuity.

`AtomisticFrameCollection` already stores one immutable three-axis PBC vector, so periodic-axis continuity is guaranteed by normalized collection construction and is recorded in the gauge signature. A future source model with frame-dependent PBC must pass an explicit continuity audit before entering this module.

Trajectory and ensemble comparison policies

Time-ordered trajectories compare frame t with the gauged frame $t - 1$. Independent ensembles compare every frame with gauged frame zero. The ensemble policy avoids inventing a temporal order.

The direct reported-basis change is

$$r_{\text{direct}} = \frac{\|H_t - \tilde{H}_{\text{cmp}}\|_F}{\max(\|\tilde{H}_{\text{cmp}}\|_F, \epsilon)}.$$

If this value is below the continuity tolerance, the reported basis is retained.

Candidate unimodular reconciliation

When the direct change is too large, form

$$A_t = \tilde{H}_{\text{cmp}} H_t^{-1}$$

and round it to the nearest integer matrix Q_t . The candidate is accepted only when:

- $|\det Q_t| = 1$;
- the normalized distance from A_t to Q_t is below `integer_matrix_tolerance`;
- the gauged cell $Q_t H_t$ is continuous with the comparison cell; and
- the gauged residual is below the stricter reconciliation residual tolerance.

The integer tolerance is intentionally larger than floating-point roundoff because small physical strain can accompany a discrete basis relabeling. The independent gauged-cell residual prevents an arbitrary affine change from being accepted only because its inverse happens to lie near an integer matrix.

If a valid nontrivial candidate is found while reconciliation is disabled, the module raises `UnsupportedBasisChangeError`. When reconciliation is enabled, the matrix is applied explicitly and retained in provenance.

A determinant-sign change may be reconciled only through a certified orientation-reversing unimodular matrix. Otherwise it raises `CellHandednessError`.

Unresolved changes

If neither the reported basis nor a certified unimodular gauge provides a continuous cell identity, the module raises `LatticeBasisContinuityError`. The first implementation does not guess whether the discontinuity is a restart, large physical deformation, malformed cell, or unsupported basis operation. Callers must segment the trajectory or provide a later explicit policy.

Persistent records

`PeriodicLatticeGauge` retains:

```
source digest
frame semantics
periodic axes
common gauged handedness
per-frame comparison frame
reported and gauged cells
integer gauge matrix
reported determinant and condition number
direct and gauged relative changes
integer-candidate residual
frame status
options and immutable signature
```

Per-frame statuses are:

reference
continuous_reported_basis
reconciled_unimodular_basis

The source digest binds frame IDs, PBC, cells, origins, and frame semantics. A selected-source-frame reference cell must carry the lattice-gauge signature.

Initial limitations

Stage C0A1 deliberately does not:

- optimize over a broad family of integer matrices;
- reconcile non-unimodular supercell changes;
- infer atom remapping after a supercell transformation;
- accept abrupt unresolved deformation as physical by default;
- segment restarts automatically; or
- construct registered coordinates.

These boundaries keep the first gauge deterministic and fail closed.

Validation

Focused tests must include:

- smooth physical cell changes;
- exact and strained unimodular shear relabelings;
- orientation-reversing unimodular relabeling;
- default rejection when reconciliation is disabled;
- explicit reconciliation with the expected inverse matrix;
- unresolved noninteger cell jumps;
- handedness changes without lattice equivalence;
- trajectory versus ensemble comparison anchors; and
- deterministic gauge signatures.